ODD-RUN REVERSAL ON CYCLIC BINARY WORDS: EXACT RECURRENCE AND SHARP PARITY-DEPENDENT TRANSIENTS

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ABSTRACT. On labelled cyclic binary words of length n, consider the parallel rule that flips every bit belonging to a maximal run of odd length. A boundary survives one round exactly when its two incident run lengths have the same parity; no new boundary is created. We prove that a word is recurrent exactly when all its cyclic run lengths have one parity. All eventual periods are therefore at most two. For odd n, the two constant words form the only recurrent orbit. For $n = 2m$, the fixed-state count is $2^{m+1} - 2$, while the number of period-two states is

$$p_{2m} = \sum_{\substack{2 \le r \le 2m \\ r \text{ even}}} \frac{4m}{r} \binom{m + r/2 - 1}{r - 1}.$$

The main temporal result is the sharp maximum preperiod

$$\max \text{depth} = \begin{cases} (n-1)/2, & n \text{ odd}, \\ \lfloor (n-2)/4 \rfloor, & n \text{ even}. \end{cases}$$

The odd case follows from run coalescence. The even case needs a different coordinate: label surviving boundaries by their site parity, delete a label when its two neighbours differ, and use a realization cost that drops by at least four at every mixed step. Binary-run enumeration, shrinking-cell models, cyclic compositions, and zeta bookkeeping receive zero contribution credit. The residual scope is only this rule-specific conjunction. Ownership, priority, and external circulation remain on hold.

1. INTRODUCTION

A local update stated in terms of maximal runs is not a fixed-radius cellular automaton: a decision can depend on the distance to the next change of bit. It nevertheless has a simple lower-dimensional trace. The rule studied here flips every odd run simultaneously. Instead of tracking the bits, we track the boundaries between runs. They can disappear but never appear.

This places the mechanism near shrinking cellular automata, where deleted cells make their surviving neighbours adjacent [3–5]. Those models and their language-recognition theory are background, not claims of this paper. Static enumeration of binary runs and its connection with compositions are also mature; a current systematic account is [2]. We give all such run and composition machinery zero contribution credit.

After that subtraction, four exact statements remain for one specified finite self-map. First, boundary survival is governed only by the parities of the two incident run lengths. Second, this gives a complete recurrent classification and labelled census. Third, elementary coalescence proves a sharp clock at odd circumference. Fourth, even circumference carries an intrinsic parity label on each boundary; its induced eroder and a cost-drop lemma prove a different sharp clock.

The parity distinction is essential. Counting boundaries gives the exact odd bound but loses a factor of two in the even case. Conversely, boundary site parity is globally well defined only when the circumference is even. The two proof routes therefore meet at the survival lemma and then separate.

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We count labelled words, not rotation classes. We do not determine every basin layer or claim that the update is identical to a named shrinking automaton. A bounded owner search did not expose the same temporal conjunction, but a search miss is neither novelty evidence nor a priority certificate. External dissemination remains **HOLD**.

2. The update and its boundary trace

Let $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, with $n \geq 1$, and let $\mathcal{W}_n = \{0,1\}^{\mathbb{Z}_n}$. A *cyclic run* is a maximal cyclic interval on which a word is constant. A constant word has one run, of length n.

Definition 2.1 (odd-run reversal)**.** For $w \in \mathcal{W}_n$, the word $\mathcal{F}_n(w)$ is obtained by flipping every bit in every odd-length cyclic run of w, simultaneously, and retaining every bit in an even-length run.

For a nonconstant word, let $\mathcal{B}(w) \subseteq \mathbb{Z}_n$ be its set of run starts: $i \in \mathcal{B}(w)$ when $w(i-1) \neq w(i)$. In cyclic order write the run lengths as $\ell_0, \dots, \ell_{r-1}$. Here $r = |\mathcal{B}(w)|$ is even.

Lemma 2.2 (boundary survival)**.** *No new run boundary is created. The boundary between consecutive runs of lengths ℓ_{i-1} and ℓ_i survives one update if and only if*

$$\ell_{i-1} \equiv \ell_i \pmod 2.$$

Proof. Every bit of one old run receives the same flip decision, so no boundary can appear inside it. Let the bit on the left of an old boundary be a, and the bit on the right be $1-a$. Put $\epsilon = \ell_{i-1} \bmod 2$ and $\eta = \ell_i \bmod 2$. The two new endpoint bits are

$$a + \epsilon, \qquad 1 - a + \eta \quad \text{in } \mathbb{F}_2.$$

They remain different exactly when $\epsilon = \eta$. □

Theorem 2.3 (recurrent classification)**.** *A word is recurrent under $\mathcal{F}_n$ if and only if all of its cyclic run lengths have the same parity. In that case it is fixed when all run lengths are even and has exact period two when all run lengths are odd. Every orbit has eventual period at most two.*

Proof. By lemma 2.2, boundary sets decrease under inclusion. If an orbit returns to a previous word, its boundary set cannot have decreased anywhere along the intervening segment. Every boundary therefore survives the first round of that segment. Equality of the two incident parities at every boundary propagates around the cyclic run list, so all run lengths have one parity.

Conversely, if all runs are even, no bit flips and the word is fixed. If all runs are odd, every bit flips. The resulting complement has the same run decomposition, so the next update returns to the original word. A binary word never equals its bitwise complement, hence this period is exactly two. Every finite orbit eventually becomes recurrent, which proves the last claim. □

3. Exact recurrent census

Let f_n and p_n denote the numbers of fixed and exact-period-two states, respectively.

Theorem 3.1 (labelled recurrent counts)**.** *For odd n,*

$$f_n = 0, \qquad p_n = 2.$$

For $n = 2m$,

$$f_{2m} = 2^{m+1} - 2, \tag{1}$$

$$p_{2m} = \sum_{\substack{2 \leq r \leq 2m \\ r \text{ even}}} \frac{4m}{r} \binom{m + r/2 - 1}{r - 1}. \tag{2}$$

TABLE 1. Exact recurrent census and sharp maximum preperiod for small orders. The table is illustrative; the formulas hold for every n.

n	fixed states	period-two states	two-cycles	max preperiod
1	0	2	1	0
2	2	2	1	0
3	0	2	1	1
4	6	10	5	0
5	0	2	1	2
6	14	32	16	1
7	0	2	1	3
8	30	90	45	1
9	0	2	1	4
10	62	242	121	2

Proof. A nonconstant binary cyclic word has an even number of runs. If n is odd, neither a sum of even run lengths nor a sum of an even number of odd run lengths can equal n. Thus only the two constant words recur. Their single run has odd length, so they form a two-cycle.

Now let $n = 2m$. For a nonconstant fixed word, all boundary gaps are even. Equivalently, its nonempty boundary set is an even-cardinality subset of one of the two parity classes in $\mathbb{Z}_{2m}$. Each parity class has m sites and $2^{m-1} - 1$ nonempty even subsets. Each boundary set supports two complementary words. Adding the two constant words gives

$$2 \cdot 2(2^{m-1} - 1) + 2 = 2^{m+1} - 2.$$

For an exact-period-two word with r runs, every run length is positive and odd. Writing $\ell_i = 2x_i + 1$, stars and bars gives

$$\#\{(\ell_0, \dots, \ell_{r-1}) : \ell_i \text{ odd}, \sum_i \ell_i = 2m\} = \binom{m + r/2 - 1}{r - 1}.$$

Choose a labelled distinguished boundary in $2m$ ways and the bit after it in two ways, then divide by the r possible distinguished boundaries of the same word. Summing over even r proves (2). ☐

For a finite map F, write

$$\zeta_F(z) = \exp\left(\sum_{j\geq 1} \frac{|\operatorname{Fix}(F^j)|}{j} z^j\right).$$

This periodic-point construction is standard [1].

Corollary 3.2 (finite zeta function)**.**

$$\zeta_{\mathcal{F}_n}(z) = (1 - z)^{-f_n}(1 - z^2)^{-p_n/2},$$

with f_n, p_n from theorem 3.1.

Proof. Theorem 2.3 leaves only fixed orbits and two-cycles. The formula is the routine product of one factor for every primitive cycle. ☐

4. The sharp odd clock

Define the preperiod

$$\operatorname{depth}(w) = \min\{t \geq 0 : \mathcal{F}_n^t(w) \text{ is recurrent}\}.$$

Theorem 4.1 (odd circumference)**.** *If n is odd, then*

$$\max_{w\in\mathcal{W}_n} \operatorname{depth}(w) = \frac{n-1}{2}.$$

Proof. At a nonrecurrent state, some adjacent run parities differ. The number of deleted cyclic boundaries is even and positive, so at least two boundaries disappear. An odd cyclic word has at most $n-1$ boundaries: a boundary at every site would be an alternating cycle, which exists only at even circumference. Monotone boundary loss therefore gives $\operatorname{depth}(w) \le (n-1)/2$.

For sharpness, write $n = 2t+1$. Use the cyclic run composition consisting of one part of length two and $2t-1$ parts of length one. The unique even run has an odd run on each side. At the next update both incident boundaries disappear, and these three runs coalesce into one even run. Every other boundary, lying between two odd runs, survives. The same description repeats with two fewer odd singleton runs. Exactly two boundaries disappear per round until a constant word remains after t rounds. For $n=1$, the constant word is already recurrent and the same formula gives zero. □

5. The parity eroder and the sharp even clock

Assume throughout this section that n is even. If a nonconstant word has surviving boundaries

$$b_0, b_1, \dots, b_{r-1} \in \mathbb{Z}_n$$

in cyclic order, define its *boundary-parity word*

$$q_i = b_i \bmod 2.$$

This label is intrinsic because reduction modulo two is well defined on $\mathbb{Z}_n$ exactly when n is even. The run between b_i and b_{i+1} has parity $q_i + q_{i+1}$ in $\mathbb{F}_2$. Applying lemma 2.2 at b_i yields the following autonomous rule.

Definition 5.1 (boundary-parity eroder)**.** For a cyclic binary word q, let Dq be the subsequence that retains position i precisely when

$$q_{i-1} = q_{i+1}.$$

The cyclic order of retained positions is unchanged.

Thus the boundary-parity word after one $\mathcal{F}_n$-update is exactly Dq. The recurrent parity words are constant and alternating: they correspond, respectively, to all-even and all-odd original run lengths.

For a nonempty cyclic word q, let

$$e(q) = |\{i : q_i = q_{i+1}\}|, \qquad C(q) = |q| + e(q).$$

Put $C(\varnothing) = 0$.

Lemma 5.2 (realization cost)**.** *If q is the boundary-parity word of a cyclic word of circumference n, then $n \ge C(q)$. Conversely, q is realizable at circumference $C(q)$, and at every circumference $C(q)+2j$, $j \ge 0$.*

Proof. The positive gap from b_i to b_{i+1} is even when $q_i = q_{i+1}$, so it costs at least two, and is odd otherwise, so it costs at least one. Summing the gap minima gives

$$2e(q) + (|q| - e(q)) = C(q).$$

Taking every gap at its minimum realizes equality. Adding two to any one gap preserves every boundary parity and realizes all larger circumferences of the same parity. Since a binary cyclic word has an even number of boundaries, alternating bit values can be assigned consistently between the chosen positions. □

Lemma 5.3 (four-unit cost drop)**.** *If q is neither constant nor alternating and has even length, then*

$$C(Dq) \le C(q) - 4.$$

Proof. Decompose q into cyclic constant runs. A singleton run survives. A run of length two disappears. A run of length at least three loses its two endpoints and retains its interior.

Let a be the number of nonsingleton runs and $h \le a$ the number of length-two runs. If s is the number of transitions of q, then $e(q) = |q| - s$, so $C(q) = 2|q| - s$. The update deletes exactly $2a$ symbols. Only a disappearing length-two run can merge its two neighbouring surviving runs; deleting one such run lowers the transition count by at most two. Thus, with s' the transition count after deletion,

$$|Dq| = |q| - 2a, \qquad s' \ge s - 2h.$$

Consequently

$$C(q) - C(Dq) = 4a + s' - s \ge 4a - 2h.$$

If $a \ge 2$, the right side is at least $2a \ge 4$.

It remains to consider $a = 1$. All other constant runs are singletons. Both the total length and the number of cyclic runs are even, so the unique nonsingleton run has odd length, necessarily at least three. It remains nonempty, no run disappears, and $s' = s$. The cost drop is then exactly four. □

Theorem 5.4 (even circumference)**.** *If n is even, then*

$$\max_{w \in \mathcal{W}_n} \operatorname{depth}(w) = \left\lfloor \frac{n-2}{4} \right\rfloor.$$

Proof. A mixed even-length parity word has at least four symbols. Its number of equal adjacencies is positive and even, hence at least two. Therefore its cost is at least six. If an original word has preperiod $t \ge 1$, its parity words before rounds $0, \dots, t-1$ are mixed. Applying lemma 5.3 for the first $t-1$ transitions and then lemma 5.2 gives

$$n \ge 6 + 4(t-1) = 4t + 2.$$

Thus $t \le \lfloor (n-2)/4 \rfloor$.

For sharpness, fix $t \ge 1$ and take

$$q = 0^{2t+1}1.$$

Its cost is $4t + 2$. Each eroder round removes the two endpoints of the long zero run. After exactly t rounds it becomes the alternating word 01, and no earlier state is recurrent. By lemma 5.2, this boundary word is realized by an original cyclic word at circumference $4t + 2$; adding two to one gap realizes circumference $4t + 4$ with the same parity dynamics. These are exactly the two even circumferences for which $\lfloor (n-2)/4 \rfloor = t$. At $n = 2, 4$, the claimed maximum is zero and the recurrent classification verifies it directly. □

6. Exact controls, scope, and conclusion

The accompanying deterministic verifier exhausts all 2^n labelled words for $1 \le n \le 16$. It compares literal updates with run data, checks the recurrent criterion, exact census, both maximum-depth formulas, and an extremal witness at every order. A fresh run executes 262,188 exact assertions. This finite calculation is falsification and regression evidence; none of the all-n statements depends on it.

The result is deliberately narrow. It gives no complete transient-layer distribution and no orbit census modulo rotation. Shrinking automata, binary-run enumeration, cyclic compositions, and periodic-point zeta bookkeeping are prior tools and receive zero credit. What remains is the exact conjunction for odd-run reversal: a boundary survival law, complete recurrence, a labelled recurrent census, and two parity-sensitive sharp clocks. Owner clearance, priority, and external release remain on hold.

References

[1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.

[2] Félix Balado and Guénolé C. M. Silvestre. Systematic enumeration of fundamental quantities involving runs in binary strings, 2026. URL https://arxiv.org/abs/2602.10005.

[3] Martin Kutrib, Andreas Malcher, and Matthias Wendlandt. Shrinking one-way cellular automata. *Natural Computing*, 16(3):383–396, 2017. doi: 10.1007/s11047-016-9588-8.

[4] Augusto Modanese and Thomas Worsch. Shrinking and expanding cellular automata. In *Cellular Automata and Discrete Complex Systems*, volume 9664 of *Lecture Notes in Computer Science*, pages 159–169. Springer, 2016. doi: 10.1007/978-3-319-39300-1_13.

[5] Azriel Rosenfeld, Angela Y. Wu, and Tsvi Dubitzki. Fast language acceptance by shrinking cellular automata. *Information Sciences*, 30(1):47–53, 1983. doi: 10.1016/0020-0255(83)90045-2.